



**Carnegie
Mellon
University**

Geopolitics of Innovation

**84-375
84-675**

WEH 4708

Monday & Wednesday 2:00 - 3:20pm

Welcome to the course!

- Introductions
- Syllabus review
- Motivation of the course

Introductions

Name (preferred nickname, pronouns if comfortable sharing)

Year/Cohort

Major (declared or intended)

Favorite class so far and why

Any clubs, labs, campus activities?

What drew you to this class (“fulfills credit” is a fine answer too)

The classroom as digital detox

After today, this will generally be a **device-free period** (exceptions noted).

During class, focus on writing down points you think are most important.

- **You are not a human dictation device.** Don't try to record verbatim.
- Handwritten notes force you to **triage** the information being conveyed—which requires you to actually think about the material!
- Internalize by **retyping** / **recopying** your handwritten notes after class.

I will share the slides on Canvas after each unit (before exams).

To ensure free expression, digital recordings (photo, video, audio) of any kind are NOT allowed. If you need accommodations, work through ODR.

Reminder on Generative AI (from Heinz College)

You are welcome to use generative AI programs (ChatGPT, DALL-E, etc.) in this class. These programs can be powerful tools for learning and other productive pursuits, including completion of some assignments in less time, helping you generate new ideas, or serving as a personalized learning tool.

However, your responsibilities as a student remain the same. **You must follow the academic integrity guidelines of the university and of this class.** If you use one of these generative AI tools to develop content for an assignment, you are required to cite the tool's contribution to your work. In practice, cutting and pasting content from any source without citation is plagiarism. Likewise, paraphrasing content from a generative AI without citation is plagiarism. Similarly, using any generative AI tool without appropriate acknowledgement will be treated as plagiarism. The [university's policy on plagiarism](#) applies to all uncited or improperly cited use of work, whether that work is created by human beings alone or in collaboration with a generative AI.

- If you provide minimum-effort prompts, you'll get low quality results. Do the work to refine your prompts.
- Don't trust it! Assume there are errors. Fact-check the output and then edit/improve it. Add in references to the scientific evidence we cover in class. This will work best for topics you understand (i.e., it's hard to use for things you know nothing about.) Ultimately, you are responsible for the accuracy of your work.

Syllabus

Theme

Learning objectives

Expectations

Weekly topics

Why this course?

OPINION
GUEST ESSAY

No Substitute for Victory

America's Competition With China Must Be Won, Not Managed

MATT POTTINGER AND MIKE GALLAGHER

May/June 2024 *Published on April 10, 2024*

In the Future, China
Will Be Dominant. The
U.S. Will Be Irrelevant.

May 19, 2025

***Europe Feels Squeeze as Tech
Competition Heats Up Between U.S.
and China***

As the rapid pace of change mixes with national security issues, Europe's role as a global regulator is increasingly tested — and may not be enough.

Why does technology matter?

Why does technology matter? Two reasons:

F. Scott Fitzgerald: “The rich are different from you and me.” Hemingway: “Yes, they have more money.” But how did rich countries come to have more money than poor countries?

- ^ See Joel Mokyr: “Lever of riches”

National power derives (in large part) from the **stock of technology**. Military/defense applications, specifically—but not exclusively. Superadditive vs. zero-sum aspects.

- ^ See Amy Zegart: “Knowledge power”

So if technology is so important, *how do countries get it? Why are some countries better at getting it than others? Is that more a matter of strategy or accident?*

What is “technology?”

- Pens down and think for a second...



14 th Century	15 th Century	16 th Century	17 th Century	18 th Century	19 th Century	20 th Century	21 st Century
Infantry Revolution - Longbow - Crossbow - Siege Warfare	Artillery Revolution - Gunpowder - Steel Revolution of Sail and Shot - Mounted Guns	Fortifications Trace Italienne	Dutch-Swedish Tactical Reforms - Combined Pike - Musketeers - Cavalry - Artillery French Military Reforms Naval Warfare (Ship of the Line)	British Financial Revolution French Revolution Industrial Revolution - Steam Power Napoleonic Revolution - Levée en Mass Professional Staff Personnel, Procedures	 American Civil War Railroads / Rifles / Telegraph	World War I Armored Warfare Battlefleet Warfare Tactical Fires Fortifications (Trenches) Medical Chemical Weapons Interwar Revolution <i>Irish Civil War</i> <i>World War II</i> Combined Arms Tactics and Operations - Close Air Support - Radar, Signals Intelligence - Motorization, Blitzkrieg Amphibious Warfare Carrier Warfare Submarine Warfare Air Defense Strategic Bombing Helicopter Airmobility Total War	Nuclear Revolution Airland Battle - Follow-on-Forces Attack Unconventional Warfare Counterterrorism Population-based Counterinsurgency (COIN) Transformation, Effects-Based Operations Targeting Revolution - Guided Missiles Information Revolution - Information / - Electronics Warfare - Network-Centric Warfare - Computers Remotely Piloted Vehicles 4 th Generation Warfare

LEGEND
Military Revolutions Innovation cases <i>Were treated as cases</i>

Number of Citations

20
15
10
5
0











Compression Algorithm

In subject area: [Computer Science](#)

A compression algorithm is a technique used in computer science to reduce the size of a file without removing information. It achieves this by finding a formula that describes the data and produces compressed data that appears more random. This is useful for various purposes such as handling less data, mimicking data, and identifying noise. Different techniques like probability methods, dictionary methods, run-length encoding, wave methods, fractal methods, and adaptive compression schemes are used in specific domains.

Underlying organizational processes?



Etymology of the term “technology”

“Technology” is a rather new word (post-1920, Heidegger) ... as is “scientist” (1834, Hewell).

Indo-European word “*tek*” → Greek “*techne*” (creation through know-how)

- Aristotle: *techne* (know-how or craft) vs. *episteme* (causal understanding)

French “technique” and German “technik” in 19th century (way of doing)

Evolves into “technologie,” from “techne + *logos*” (word, discourse, or study).

- 1700s: Education denotation (Christian Wolff, Joseph Beckman, Francis Bacon, René Descartes)
- 1800s: Morphed to “technology” in American English (“Carnegie Institute of *Technology*”)
- Prior to late 19th century, “science” refs. to a body of knowledge, not a profession or method.

Resurgence of “technology” in 1920s as a substitute for “inventions” (**modern switchpoint**)

Meaning continues to evolve.

Oxford English Dictionary	• 1 : The application of scientific knowledge for practical purposes, especially in industry. • 2 : machinery and equipment developed from the application of scientific knowledge. • 3 : the branch of knowledge dealing with engineering or applied sciences.
Britannica	• 4 : The application of scientific knowledge to the practical aims of human life • 5 : or, as it is sometimes phrased, to the change and manipulation of the human environment.
Merriam - Webster	• 6 : the practical application of knowledge especially in a particular area • 7 : a capability given by the practical application of knowledge • 8 : a manner of accomplishing a task especially using technical processes, methods, or knowledge • 9 : the specialized aspects of a particular field of endeavor
Wikipedia	• 10: Technology is the application of conceptual knowledge to achieve practical goals, especially in a reproducible way.[1] • 11 : The word technology can also mean the products resulting from such efforts,[2][3] including both tangible tools such as utensils or machines, and intangible ones such as software.

Exercise: convergence/cleavages in common definitions

Practical Ability

Human Creativity

Effort and Intention

Organized Knowledge

Solution-Oriented

Tangible

Includes *Practices*

Modifies the Natural

Improves

Several popular schools of thought

1. Ferré's Axioms
2. Joseph Schumpeter "creative destruction"
3. Martin Heidegger (1954)
4. Arthur's Three Criteria
5. Kim Clark & Rebecca Henderson "architectural innovation"

1. Ferré's 3 Rules - tech as know-how *going somewhere*



Technology is embodied, not in the head alone.

Technology is practical and is implemented.

- Not “for its own sake”; not “empty-handed”

Technology is intelligent, not blind.

- Must have some degree of calculation about getting from means to end.

2. Arthur's 3 Criteria - “**Recombinatory Technologies**”

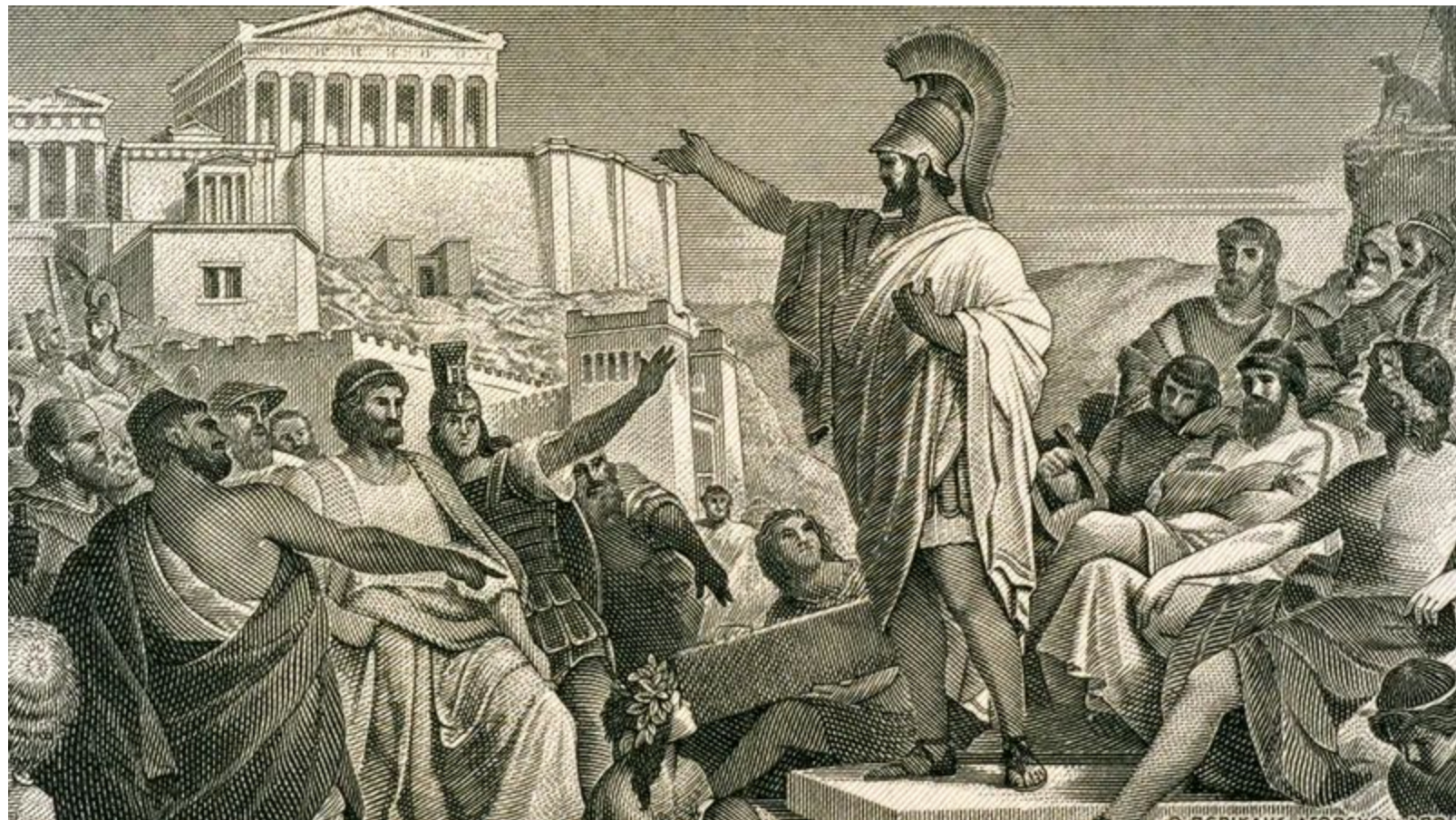
Technology “works” by exploiting natural phenomena. Technology is not just a *means*; it is the ***programming of phenomena for a purpose***.

1. “**Effect**”: does it harness a natural phenomenon to change the environment?
2. **Combinatory**: “...[a] technology as an **assemblage of practices and components**” and can also be in a broader sense “... the entire collection of devices and engineering practices available to a culture.”
3. **Recursive**: Each technology stands ready to become a component in another technology. **Nested**.

Can animals have technology?









PROPHET OF INNOVATION



Joseph Schumpeter and Creative Destruction

THOMAS K. McCRAW

WINNER OF THE PULITZER PRIZE

Why does technology **change**?

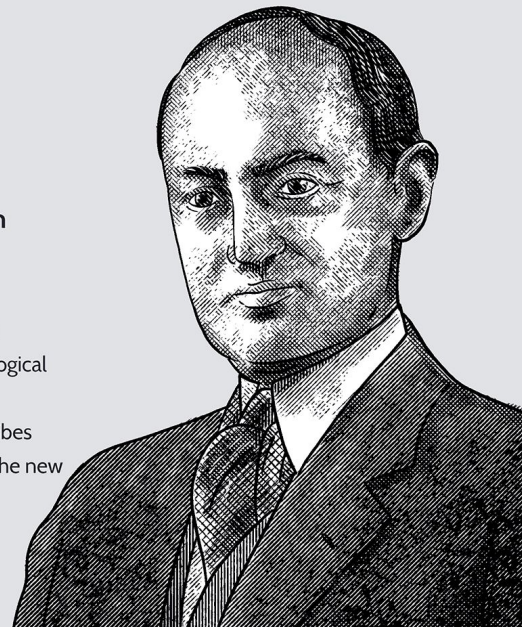
Joseph Schumpeter

Born: February 8, 1883

Died: January 8, 1950

Economist and Economist Historian

- Best known for his 1942 book "Capitalism, Socialism, and Democracy," the theory of creative destruction, and for offering the first German and English references to methodological individualism in economics
- Creative destruction, a term he coined, describes how the old is constantly being replaced by the new



 Investopedia

Heidegger (1954)

Common view: People think technology is a means to an end and a human activity.

- “Instrumental” (or “anthropological” view)

Technically correct, but **insufficient** for a definition. NOT just a neutral tool.

More important aspect of technology: **it changes our relationship to the world.**

- Invention of the windmill was (in an immediate sense) about harnessing cheap energy, but actually it helped us understand how wind works! Technology reveals new truths about reality.
- Coal mine or nuclear plant acts as a DEMAND on nature, forces it to give up something it wouldn't have naturally.

Invention of telescope didn't just improve our ability to identify distant ships. It helped us realize our place in the universe! Its inventor didn't really “intend” that.

- Purpose is ancillary, or even (in Heidegger's opinion) a dangerous distraction.

Hand in homework assignment (tech definitions crosstab)

Typologizing technology

Tangibility: Hardware (material) ↔ Software (practice)

Hierarchy: Components (part) ↔ Platform ↔ System (whole)

Design: Incremental ↔ Radical

Application: Single-Use ↔ Dual-Use

Scope: Enabling Technologies, General-Purpose Technologies (GPTs), Systemic, etc.

Technology Presentation Signups

13 students total, teams of 3 each = 4 teams present over 2 days.

- 25 minute presentations on a technology of your choosing.
- Presentation instructions in the syllabus (assignments section)
- Dates: **September 28 & 30**
- Teams that volunteer for 28th get first pick topic
- Please email me with (1) **your team's preferred date** and (2) **top-3 technology picks** (all members cc'd) by *this Friday*.

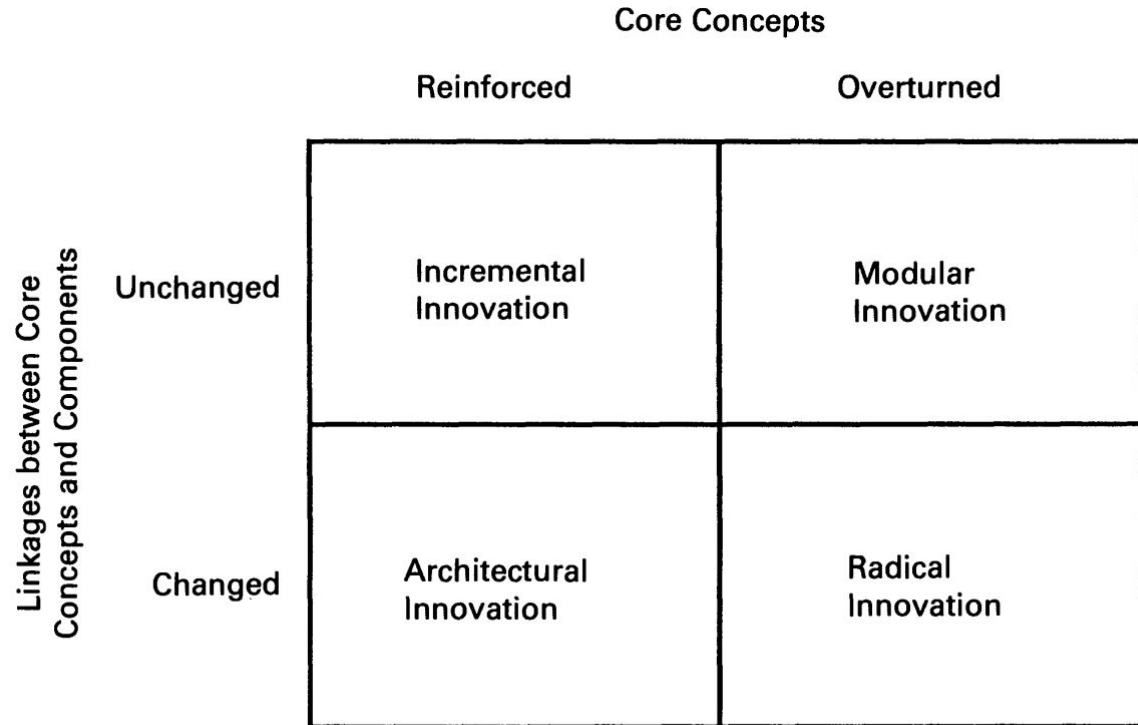
Know-how is key



Know-how is key



Clark & Henderson (1990)



Clark & Henderson (1990)

		Core Concepts	
		Reinforced	Overturned
Linkages between Core Concepts and Components	Unchanged	Incremental Innovation	Modular Innovation
	Changed	Architectural Innovation	Radical Innovation

Recapping lesson

1. Technology on a **spectrum** [typology]
 - a. How we define technology has implications for strategies to improve it
2. “Know-how” about how to harness nature, impact the environment
 - a. Purpose: know-how about how to create/produce “artificial” effects
 - b. Practical: know-how about how to use what we’ve created
 - c. *Scientific* knowledge is not always a prerequisite, even if science has ability to explain it.
3. **Cumulative: chain of know-how**, even if antecedent links are lost
 - a. Ferré’s Rules (**intelligent progression**)
 - b. Arthur’s Criteria (**radical** OR **recombinant** - two kinds of “novelty”)
 - c. Always borrows from existing links; no such thing as wholly radical impact (technology vs. magic)
4. Clark & Henderson (**design AND architecture** - both matter)
 - a. Always (?) manifested in *something* tangible, but need not be any core component.
 - b. Offer a compromise in the **debate over tangibility** (embodiment != tangibility).
5. “*Technological*” + “*innovation*”: **arc** is improvement, but not every tech improves.
6. ^ **Necessary but not sufficient** criteria. (ie. exclude institutions?)

Activity: Can we come up with a better definition?



Prompt: *“How are technologies different from customs or institutions?”*

Keep pushing the LLM until it gives you a criteria set that exclude things like legal systems but still include (mostly) immaterial “technologies” (eg. blockchain ledgers).

You may use any LLM you like.

We will vote on the best definition.

<https://forms.gle/H3BVV3JZYSZni3pq9>

Next lesson: technology genesis.